

Weekly Report 2

Automated Biometric System for Individual Re-Identification of Mugger Crocodile

Siya Dhaduk (AU2340022)
Rena Malhotra (AU2340018)
Shayan Pariyal (AU2340107)
Kaushika Rathod (AU2340133)

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1 Introduction

This project aims at developing an automated system that will be capable of recognizing individual Mugger crocodiles from images. The system has to identify whether a crocodile in a new image is already known or if it is a new individual. This is significant since the survival of individual Mugger crocodiles can be monitored in terms of population dynamics, movement and survival throughout time.

In this phase of the project, we focused on implementing baseline model for crocodile detection and classification. The work involved preparing a limited dataset, applying data augmentation techniques, and experimenting with convolutional neural network architecture for image classification and object detection.

2 Dataset Preparation

The initial data used in the article by Desai et al. [1] comprises a large amount of Mugger crocodile images taken during UAV images in various habitats in Gujarat. The dataset link that was given, however, only included test data folders, which were therefore a major hindrance, reducing the size of the available labeled images to train. Thus, in the first implementation of the baselines, we took the following data:

Therefore, for the initial baseline implementation, we used the following dataset:

- Gharial Crocodile – 16 images
- Saltwater Crocodile – 16 images
- Unknown Crocodile – 208 images

We used wide data augmentation to augment the data since it was small. The heterogeneity of training samples. The methods of augmentation were:

- Image scaling

- Image rotation
- Horizontal and vertical flipping
- Minor geometric transformations

These augmentation techniques were used to simulate the changes in aerial imagery that included a variety of viewing angles, orientations, and scales.

Class imbalance was also considered while preparing the dataset to ensure that each category contributed equally during training.

3 Baseline Model Implementation

To establish an initial baseline for crocodile detection and classification, we implemented the following deep learning approach:

3.1 Inception-v3 Based Classification

We used the pre-trained **Inception-v3 convolutional neural network** as a baseline image classification model. The network was narrowed down to categorize pictures into the following categories:

- Gharial Crocodile
- Saltwater Crocodile
- Unknown Crocodile

The final classification layer was substituted with the model architecture to fit the size of our data set in terms of classes, learning rate, and batch size. We experimentally selected the hyperparameters as of now. We will later consider the hyperparameter tuning when we get the complete dataset.

We planned to evaluate the performance of the model using the following Key Performance Indicators (KPIs):

- Accuracy
- Loss
- Precision
- Recall
- F1-score

4 Preliminary Results

The baseline models successfully performed basic crocodile classification on the augmented dataset. The given dataset has a limited number of images in test sets 1 and 2, because of which the model overfitted. Yet, we are not still sure. Currently we are getting 100% test accuracy. We think that this is due to the fact that we have a very small number of images per class and the depth of model is quite high at the same time. We tried to go with data augmentation. But after a certain limit of number of images, we are getting RAM usage error.

5 Challenges Faced

During the initial phase of the project, several challenges were encountered:

- Limited availability of labeled training images.
- Difficulty in selecting optimal hyperparameters due to limited training samples.

6 Plan for Next Week

The next phase of the project will focus on moving from simple classification toward the main goal of **individual crocodile re-identification**. The planned tasks include:

- Our first task would be to acquire complete dataset if possible. Else, find a solution for this.
- Extracting scute pattern features for individual identification.
- Implement YOLO-V5 model for the identification.

References

- [1] B. Desai, A. Patel, V. Patel, S. Shah, M. S. Raval, and R. Ghosal, "Identification of free-ranging mugger crocodiles by applying deep learning methods on UAV imagery," *Ecological Informatics*, vol. 72, p. 101874, 2022.